PROTOTYPE — REVIEW 1 / 35%

SMART DUSTBIN

Prototype Design & Development Document

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Review Stage	Review 1 – 35% Project Completion

“Be smart, use smart dustbins.”

1. Prototype Overview

The Smart Dustbin prototype is a low-cost, sensor-and-AI based concept for monitoring the fill level of a public dustbin and notifying collection staff before overflow. The prototype uses an ultrasonic sensor for continuous fill-level estimation and a camera-based object-detection component for visual overflow verification. An ESP32 acts as the main processing unit, while Wi-Fi/GSM provides the communication path.

2. Prototype Goals

- Measure the approximate amount of waste inside the bin.
- Detect when the bin approaches the 80% alert threshold.
- Provide a mechanism for automatic notification.
- Use camera-based AI as an additional overflow-verification layer.
- Create a foundation for demand-based waste collection.

3. Prototype Components

Component	Role in Prototype
Smart dustbin	Physical waste container and mounting structure
Ultrasonic sensor	Measures distance to the waste surface
ESP32	Reads sensor data, processes fill level and controls logic
Camera	Captures images for visual inspection
AI object-detection model	Identifies the bin and visible spillage/overflow
Wi-Fi/GSM	Sends status or alert information
Power supply	Provides power to the prototype electronics

4. Prototype Architecture

Input / Sensing	Processing	Communication	Output
Ultrasonic Sensor Camera	ESP32 Fill-level logic AI verification	Wi-Fi / GSM	Dashboard SMS Alert Collection Action

Figure 1. High-level prototype architecture.

5. Physical Prototype Layout

The ultrasonic sensor is intended to be mounted near the top/lid of the dustbin and pointed toward the waste surface. The ESP32 and communication electronics can be housed in a protected enclosure. The camera should be positioned so that it can view the bin and the nearby area where spillage may occur.

Location	Planned Placement
Top / Lid	Ultrasonic Sensor
Front / Side	Camera
Protected Electronics Box	ESP32 + Communication
Inside Bin	Waste / Fill Area

6. Fill-Level Measurement

The ultrasonic sensor measures the empty distance between the sensor and the waste surface. If H represents the internal bin depth and D represents the measured distance from the sensor to the waste, the approximate fill percentage can be calculated as:

$$\text{Fill \%} = ((H - D) / H) \times 100$$

Example: if the usable bin depth is 100 cm and the measured empty distance is 20 cm, the estimated fill level is 80%. At the 80% threshold, the prototype can enter the alert state.

7. Alert Logic

Estimated Fill Level	Prototype Status	Action
0–50%	Normal	Continue monitoring
50–79%	Monitor	Continue monitoring
80–99%	Alert	Notify collection team
100% / visible spillage	Overflow risk	Prioritize collection

8. AI Overflow Verification

The proposed camera component captures images around the dustbin. The AI object-detection model is intended to learn from labelled images in which bounding boxes identify the bin and spilled waste. The AI output can be used as a visual confirmation layer when overflow is suspected. At the 35% stage, this is a proposed component; a trained model and measured accuracy are part of future work.

9. Prototype Working Sequence

- Start the smart dustbin and initialize the ESP32.
- Read the ultrasonic sensor at regular intervals.
- Convert the measured distance into an estimated fill percentage.
- Compare the fill percentage with the 80% threshold.
- If the threshold is crossed, trigger the communication path.
- Use the camera/AI component to verify visible overflow where applicable.
- Send a dashboard/SMS notification to the collection team.

• After collection, continue sensing without requiring a manual reset.

10. Prototype Pseudocode

```
START
Initialize ESP32
Initialize ultrasonic sensor
Initialize camera / communication
REPEAT
Read distance from ultrasonic sensor
Calculate fill percentage
IF fill percentage >= 80%
Capture/check visual condition
Send alert to collection team
ELSE
Continue monitoring
END IF
UNTIL system is stopped
END
```

11. Prototype Development Status – 35%

Prototype Area	Current Status
Problem and requirements	Defined
Component selection	Defined
System architecture	Designed
Fill-level logic	Defined
AI overflow concept	Defined
Physical assembly	Next phase
ESP32 implementation	Next phase
Camera integration	Next phase
AI model training	Next phase
Alert/dashboard integration	Next phase
Testing and validation	Next phase

12. Risks and Mitigation

Risk	Mitigation
Sensor fouling from dust, moisture or splashes	Protective housing and periodic cleaning
Incorrect sensor readings	Calibration and repeated readings
False overflow alerts	Use camera/AI confirmation as an additional signal

Risk	Mitigation
Communication failure	Monitor connectivity and retain a local status indicator where possible

13. Future Prototype Enhancements

- Complete physical hardware assembly.
- Calibrate and validate fill-level readings.
- Train the overflow object-detection model using labelled images.
- Integrate the communication and notification mechanism.
- Develop a live dashboard for multiple bins.
- Store historical fill-level data.
- Identify peak-fill periods for future route planning.
- Conduct field testing and compare results with the baseline.

14. Prototype Conclusion

The prototype establishes a practical architecture for a Smart Dustbin that combines sensing, embedded processing, communication and AI-based visual verification. The current 35% milestone demonstrates the design and technical direction of the prototype. The next stage is implementation and validation of the complete system.

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